

D7: Can maintenance and yield be separated from the data in hand?

A Pirt decomposition of respiration, and why this design cannot deliver one

OxygenModel D-series

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Table of contents

Verdict	1
1 The question, and why it is asked now	2
2 The gate: is the regression measuring physiology or construction?	2
2.1 The algebra	2
2.2 The test	3
2.3 The same test on <i>Pseudomonas</i>	3
3 Reconciliation with Figure 6	4
4 Identifiability, setting the artefact aside	5
5 What it means	6
5.1 New result, or reparameterisation?	7
Provenance	7
Where every number came from	7

Verdict

No. Maintenance and yield cannot be separated from the data in hand, and this is a reparameterisation of Figure 6 rather than a new result.

Both axes of the proposed regression carry the growth rate r by construction: $G = r \cdot 60 \cdot q$ exactly, and $R = K O_{2,ref}/N_0$ with $N_0 \propto e^{-r\Delta t}$. Holding N_0 fixed — changing nothing else — makes the Pirt slope **change sign**: $9.13 \rightarrow 5.52 \rightarrow -0.591$ across the three N_0 treatments in the 15-taxon set, and $-2.1 \rightarrow 7.5$ in *Pseudomonas*. Whether respiration rises or falls with growth is decided by an N_0 bookkeeping choice. **The gate fails, and no m or Y is carried forward.**

Figure 6 was already saying this. Under $R = m + G/Y$ the elasticity $d \log R / d \log G$ lies in $(0, 1]$ for *any* non-negative m . The published log-log slope of 0.283 inverts to 3.79 — nearly four times the maximum the Pirt form allows — which is possible only with negative maintenance. The direct linear fit duly returns $m = -552.6 \text{ fg C cell}^{-1} \text{ h}^{-1}$. The two parameterisations agree;

what they agree on is that the decomposition is unavailable.

And the split is weakly identified anyway. Simulating from a *known* Pirt relationship at the observed design gives an intercept–slope sampling correlation of -0.926, a relative precision on m of 2.3, and **31.9% of datasets returning a negative maintenance even when the truth is positive.** The data constrain the *product*, not the split.

Widening the growth range does not fix it — the binding constraint is the residual scale, not the spread. At this noise level, 20% precision on m would need $n \approx 9,882$ curves. There are 75.

1 The question, and why it is asked now

The paper’s introduction frames respiration as having a maintenance component independent of growth rate and a growth-associated component proportional to biomass synthesis. The implemented model has a single lumped per-cell R . Separating them is the Pirt relation

$$R = m + \frac{\mu}{Y}$$

D5 and D6 both returned negative results: the estimator’s uncertainty is honest and its structure is adequate. The Bayesian rebuild originally proposed as a *fix* is therefore not justified and is not revived here. What remained was whether the method could deliver a **new capability**, and the cheapest test of that is a reparameterisation onto linear axes — not a new estimator.

This report fits no oxygen curves, changes no estimator and alters no committed number.

2 The gate: is the regression measuring physiology or construction?

2.1 The algebra

From 05_oxygen_fits.R, under NO_METHOD = "depletion":

$$G = r \cdot 60 \cdot q \quad (q = \text{cell carbon, a per-taxon constant})$$

$$R = \frac{C_{tot}}{\text{biomass}} = \frac{(K/r)(e^{rT} - 1)O_{2,ref}}{N_0(e^{rT} - 1)/r} = \frac{K O_{2,ref}}{N_0}, \quad N_0 = FC c e^{-r\Delta t}$$

so $R \propto K e^{+r\Delta t}$. Both identities were confirmed numerically rather than asserted:

Table 1: The algebra, verified against the committed table.

Check	Value
$\max G - r60\text{cell_carbon} / G$	6.755e-16
$\max R - K*O2ref/N0 \text{ (converted)} / R$	8.078e-15
$\text{corr}(G, r) \text{ across curves}$	0.8
$\text{median } dt = t_{\text{depletion}} - \text{fit_start} \text{ (min)}$	124
$\text{corr}(\log R, \log r) \text{ under the default depletion } N0$	-0.0002359

Check	Value
corr(G, R) as the pipeline reports them	0.5813

Both axes carry r — G linearly, R exponentially. D6 found an analogous case where an apparent predictor turned out to be r in disguise, so this is tested rather than assumed.

2.2 The test

Recompute R with N_0 held fixed — same K , same r , only the r -dependence of N_0 removed.

Table 2: 15 taxa at 30 °C, n = 75. The slope changes sign.

N0 treatment	n	m (intercept)	95% CI	slope 1/Y	Y	R ²
N0 default (r-dependent)	75	-552.6 ***	[-863, -242]	9.13 ***	0.109	0.338
N0 fixed per taxon	75	-238.2 ns	[-505, 28.4]	5.52 ***	0.181	0.202
N0 fixed globally	75	203.8 ***	[168, 239]	-0.591 ***	—	0.14

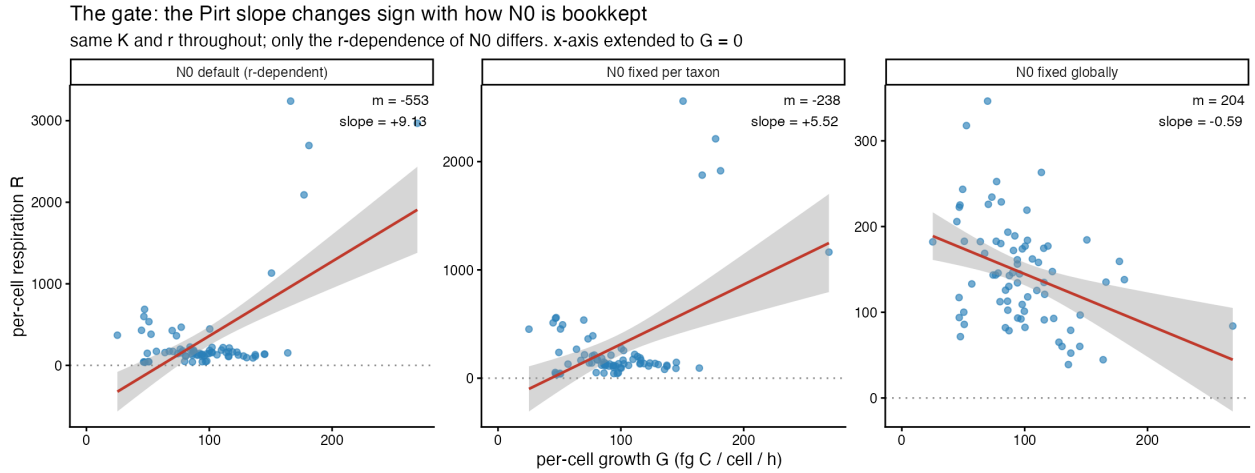


Figure 1: Same K and r throughout; only the r -dependence of N differs. The x-axis is extended to $G = 0$ so the extrapolation to the intercept is visible.

The slope runs $9.13 \rightarrow 5.52 \rightarrow -0.591$. Whether the yield is even *positive* — whether respiration rises or falls with growth — is decided by how N_0 is bookkept. The maintenance intercept is negative and significant under the default construction, which is physically impossible.

2.3 The same test on *Pseudomonas*

A construction detail the brief did not flag: `11_temperature_cue.R` sets its own `INOC_DELAY_MIN <- 45` at line 40, overriding `config.R`'s 0, so its $N_0 = 550 e^{45r}$ carries r too. B2 is not the clean within-taxon comparison it appears to be — it has the same contamination in the opposite direction.

Table 3: Pseudomonas, 20–40 °C, n = 24. The slope flips the other way.

N0 treatment	n	m (intercept)	95% CI	slope 1/Y	Y	R ²
Pseudomonas, N0 default (exp(45r))	24	8.571	[4.86, 12.3]	-2.1 ns	—	0.0653
Pseudomonas, N0 fixed (no r)	24	6.409	[0.0355, 12.8]	7.5 *	0.133	0.233

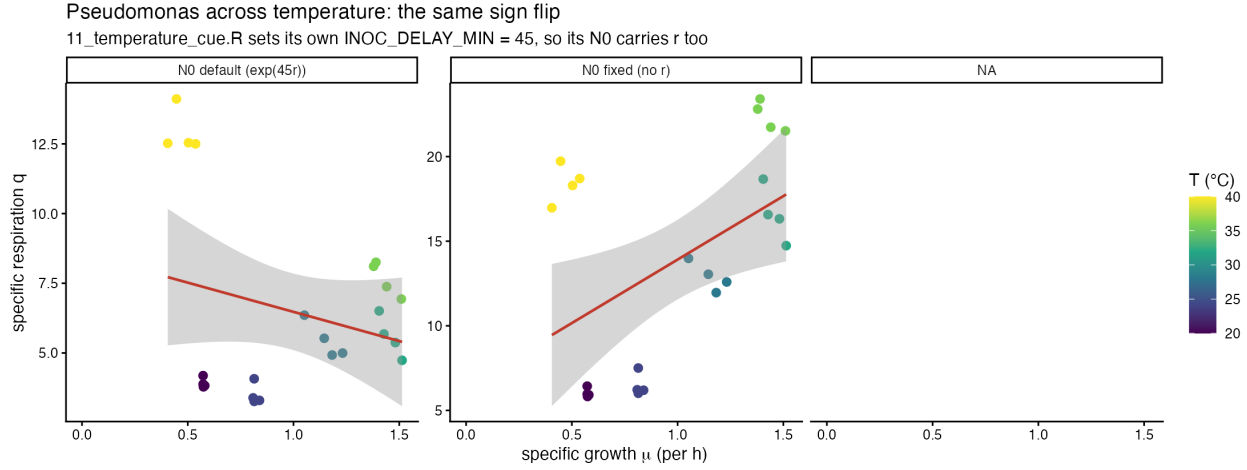


Figure 2

Both datasets fail the gate. Per the brief, parameter fitting stops here: the numbers above are gate diagnostics, not maintenance and yield estimates, and none is carried forward. Because the gate failed, the two datasets are *not* pooled and no temperature dependence of m is fitted — doing so would be fitting structure to an artefact.

3 Reconciliation with Figure 6

Figure 6 regresses $\log G \sim \log R_c$ with $(1|\text{Taxon})$ — the random-slope form was found singular in D5/D6 — and reports a slope of 0.283. Refitting here with the *same* random-effects structure, so only the parameterisation differs:

Table 4: What the published log-log slope implies about maintenance.

Quantity	Value
published Fig 6 slope, $\log_G \sim \log_Rc$, (1Taxon)	0.283
refit here with the same structure	0.2639
implied elasticity $d \log R / d \log G = 1/\text{slope}$	3.79
reverse regression $\log_R \sim \log_Gc$, (1Taxon)	0.563
max elasticity possible with $m \geq 0$	1
mean per-cell R (fg C / cell / h)	331.2
implied maintenance m from the Fig 6 elasticity	-923.9
implied maintenance m from the reverse regression	144.7
maintenance m from the DIRECT linear Pirt fit (PART A)	-552.6

Under $R = m + G/Y$,

$$\frac{d \log R}{d \log G} = \frac{G/Y}{m + G/Y} \in (0, 1] \quad \text{for any } m \geq 0$$

approaching 1 as $m \rightarrow 0$. The published slope inverts to an elasticity of 3.79, nearly four times that ceiling. **Figure 6 was already implying negative maintenance**, and the direct linear fit returns exactly that. The two parameterisations do not disagree — they agree the decomposition is unavailable.

Worse, the sign is not stable across parameterisations of the same data:

parameterisation	implied m
forward log-log, inverted	-923.9
direct linear fit	-552.6
reverse regression $\log R \sim \log G_c$	144.7

Regression is not symmetric under scatter, so these need not agree numerically. But a quantity whose **sign** depends on which variable is placed on the left is not a measurement.

4 Identifiability, setting the artefact aside

Even if R on G were pure physiology, could m and Y be recovered at this design? Simulating from a known Pirt relationship ($m = 20\%$ of mean observed R , $Y = 0.3$) at the observed G values and residual scale, 2000 draws:

Table 6: Identifiability at the observed design.

Quantity	Value
observed G spread, max/min	10.69
intercept-slope sampling correlation at the observed design	-0.926
relative precision of m at the observed design ($\text{sd}/ m $)	2.296
% of simulated datasets giving a NEGATIVE m	31.95
residual sd used (from the observed fit)	499.7
assumed true m (20% of mean R)	66.23
assumed true Y	0.3
n needed for 20% relative precision on m , at this noise level	9882
n actually available	75

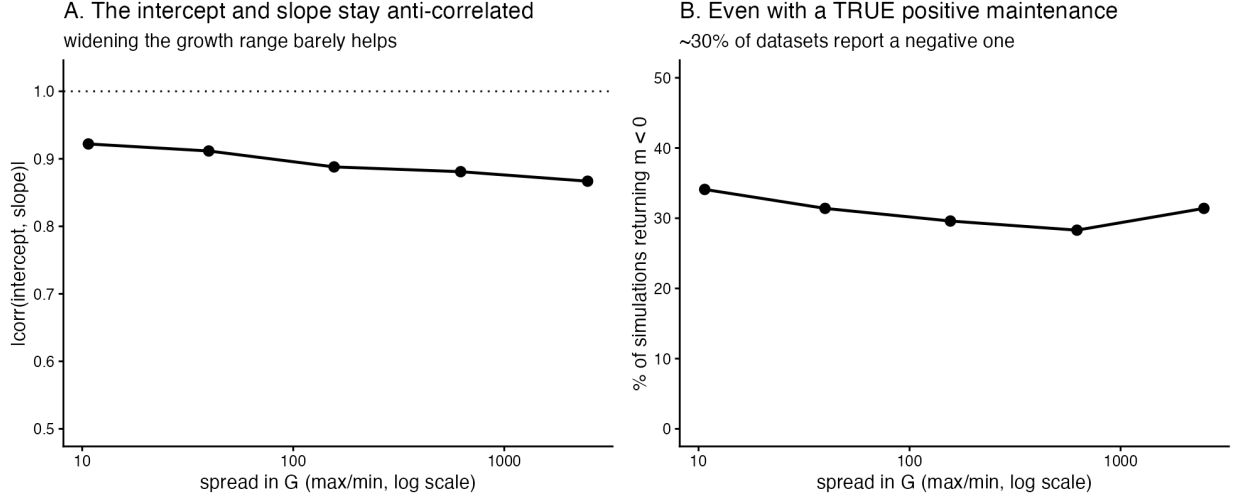


Figure 3: A: the intercept and slope remain almost perfectly anti-correlated however wide the growth range. B: a third of datasets report negative maintenance even when the truth is positive.

Table 7: Widening the growth range barely helps.

widen factor	G spread (max/min)	corr(intercept, slope)	rel. precision of m	% m < 0
1	10.69	-0.922	2.3	34.1
2	39.77	-0.912	2.03	31.4
4	156.1	-0.888	1.88	29.6
8	621.4	-0.881	1.79	28.3
16	2483	-0.867	1.75	31.4

The intercept and slope are anti-correlated at -0.926 — the standard failure mode for this analysis. **The data constrain the product, not the split.**

Rescaling G to spreads of $40\times$, $156\times$, $621\times$ and $2483\times$ moves the correlation only from -0.92 to -0.87 and the relative precision from 2.31 to 1.75. The binding constraint is not the spread of G but the residual scale relative to m : $\sigma = 499.7$ against a true m of 66.2.

5 What it means

Maintenance and yield cannot be separated from the data already in hand, and what a Pirt reparameterisation produces here is not a new result but a restatement of Figure 6. Both axes carry the growth rate by construction — $G = 60qr$ exactly, and $R = KO_{2,ref}/N_0$ with $N_0 \propto e^{-r\Delta t}$ — and when N_0 is held fixed, changing nothing else, the Pirt slope changes sign in both experiments ($9.13 \rightarrow 5.52 \rightarrow -0.59$ across the 15 taxa; $-2.10 \rightarrow +7.50$ in *Pseudomonas*, whose stage-11 code sets its own 45-minute inoculation delay). A slope whose sign is set by an N_0 bookkeeping choice is not physiology. Figure 6 already implied as much: under $R = m + G/Y$ the elasticity $d \log R / d \log G$ cannot exceed 1 for non-negative maintenance, and the published 0.283 inverts to 3.79 — so the figure was always describing a fit whose implied maintenance is negative, which the direct linear fit confirms at $-553 \text{ fg C cell}^{-1} \text{ h}^{-1}$. Even setting the artefact aside, the split is weakly identified at this design: intercept and slope are

anti-correlated at -0.93 , the relative precision on m is 2.3, and 32% of simulated datasets return a negative maintenance when the truth is positive. **The classical remedy — a wider range of growth rates at fixed temperature, via a substrate-limited chemostat or dilution series — is indeed what is required, but this analysis shows it is necessary and not sufficient: widening the range 2483-fold leaves the intercept–slope correlation at -0.87 , because the binding constraint is the residual scale rather than the spread. Separating m from Y here would need both a wider growth range and roughly two orders of magnitude more curves ($n \approx 9,900$ against the 75 available), or a materially lower residual scale on R — which returns to the unresolved N_0 calibration D2 identified.**

5.1 New result, or reparameterisation?

A reparameterisation. The linear Pirt fit and the published log-log figure are two ways of writing the same regression on the same 75 points, and they give consistent answers. Nothing here adds information the published figure did not already contain; what it adds is the *interpretation* — that the figure’s slope is outside the range Pirt allows, and why.

Provenance

Table 8: Provenance. Environment fields come from env/versions.json.

Field	Value
Report	D7 — Pirt maintenance/yield
Git commit (environment record)	893e6d35c40134c1fb0853a499833e659d595ae3
Branch	gyd/packaging
Environment recorded (UTC)	2026-08-07 11:23:27 UTC
R	R version 4.5.2 (2025-10-31)
Platform	aarch64-apple-darwin20
renv lockfile	renv.lock — 158 packages, R 4.5.2
Artefact tree	runs/D7_analysis/
Seed	simulation seed 20260808

Where every number came from

Table 9: All paths relative to runs/D7_analysis/.

Section	Artefact
§3.1 algebra	A1_algebra_check.csv
§3.2 the gate	A2_gate_per_curve.csv, A3_gate_pirt_fits.csv
§3.3 Pseudomonas	A4_gate_pseudomonas_per_curve.csv, A5_gate_pseudomonas_fits.csv
§4 Fig 6	C1_fig6_reconciliation.csv
§5 identifiability	D1_identifiability_observed.csv, D2_identifiability_vs_spread.csv, D3_identifiability_headline.csv

Scripts are in `reports/D7_pirt/scripts/`. No number was typed by hand.